NFH-Kuza-Google Implementation Plan

Eight phases from mobilisation to pilot stabilisation

Kuza Delivery Team

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NFH-Kuza-Google Pilot — High-Level Implementation Plan
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NFH-Kuza-Google Pilot — High-Level Implementation Plan

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Introduction

This document sets out the high-level implementation plan for the NFH-Kuza-Google pilot. The pilot delivers a unified, domain-agnostic service-orchestration platform for rural households in Kenya. It spans two Beckn domains — `local-services` (agri / rural) and `health-services` — sharing a single Flow Engine, channel layer, data foundation, and integrations with Google Cloud services and the Beckn network.

The plan is organised into **eight phases**. Each phase has:

- a single **Goal** statement,
- a list of **Inputs Required** — named operational artifacts (lists, datasets, content, credentials, decisions) that must be provided before the phase can start, with the provider of each artifact identified, and
- a set of **Expected Outputs** — concrete deliverables produced by the phase.

The intent of this document is to surface every input that the pilot team needs from partners and from internal teams. Once the high-level plan is reviewed and approved, a detailed plan per phase will follow, breaking each phase into tasks, owners, and durations.

Phase Summary

#	Phase	One-line Goal
0	Project Mobilisation	Governance + environments
1	Foundation — Data Model & Master Data	Schema, seeds, data dictionary
2	Beckn Network Onboarding	ONIX running, BAP/BPP registered

#	Phase	One-line Goal
3	Core Platform Services	Flow Engine + BAP + BPP running
4	Channel Integration	WhatsApp Flow Handlers, Web App, Google AI live
5	Vertical Rollout — Local Services	4 agri/rural flows live
6	Vertical Rollout — Health Services	6 health flows live, cascades + batches
7	Pilot Launch	Real cohort on the system
8	Stabilisation & Analytics	KPIs, dashboards, feedback loop

Sequencing notes:

- Phase 0 unblocks everything else.
- Phases 1 and 2 can run in parallel once Phase 0 is complete.
- Phase 3 depends on Phases 1 and 2.
- Phase 4 depends on Phase 3.
- Phases 5 and 6 default to **sequential** (Local Services first, then Health Services), though they can parallelise if two independent squads are available.
- Phase 7 is gated by Phase 5 *and* Phase 6 (whichever subset is in scope for launch).
- Phase 8 begins with pilot go-live and runs continuously thereafter.

Phase 0: Project Mobilisation

Goal

Establish governance, environments, and shared understanding before any code is written.

Inputs Required

Input	Type	Provider
Pilot Charter (scope, timeline, budget, success criteria)	Document	Kuza + Google + Beckn jointly
Delivery Team Roster (names, roles, time commitments)	List	All partners
Steering Committee Members & Cadence	List + schedule	Partners
GCP Billing Account ID & owner approval	Credential	Google / Kuza
GCP Project Naming Convention (dev / staging / pilot)	Decision	Kuza tech lead
GitHub Org + Repo Naming Convention	Decision	Kuza tech lead
WhatsApp Business Account approval	Account	Meta (via Kuza)
Beckn Fabric Testnet subscriber-registration approval	Account	Beckn
Communication tooling (Slack / email channels, doc store)	Setup	Kuza
Decision Log Template	Template	Kuza

Expected Outputs

- GCP projects provisioned (dev, staging, pilot)
- Shared Git repositories with branching strategy and CI scaffolding
- Secrets management configured (GCP Secret Manager)
- Decision log live for open items (e.g., PostgreSQL migration timing, monitoring stack)
- RACI matrix for the delivery team

Phase 1: Foundation — Data Model & Master Data

Goal

Lay down the domain-agnostic data foundation that all subsequent work depends on.

Inputs Required

Input	Type	Provider
Kenya Geography Master List (47 Counties × all Sub-Counties × Wards × known Villages)	Dataset (CSV)	Kuza (existing) or KNBS
Top-level Category List per domain	List	Kuza product team
Sub-category Taxonomy per top-level category	List per category	Kuza product team + domain SMEs
Vertical Catalogue (full list of verticals to support in pilot, with `vertical` codes)	List	Kuza
`verticalAttributes` Field Definitions per vertical	Per-vertical doc	Domain SMEs
Outcome-Type Dictionary per vertical (e.g., TB: TB_POSITIVE, TB_NEGATIVE, CURED, LOST, DIED)	List per vertical	Domain SMEs
Capability Vocabulary List (GeneXpert, X-ray, Spirometry, Vet, Vaccination, Brooding, ...)	List	Kuza + health partner
Role Definitions Dictionary (Seeker, Facilitator, Provider,	List	Kuza

Input	Type	Provider
Coordinator, Validator, Sponsor + sub-roles)		
Service Status Lifecycle Enum (NEED_REGISTERED → CLOSED + branches)	Confirmation	From `/unified` spec
Currency + Language Codes Supported in Pilot	Decision	Kuza
Master Data Sign-off Matrix (who approves each list)	Matrix	Kuza PM

Expected Outputs

- ERD and DDL for new marketplace tables (seeker profile, facilitator profile, provider profile, flow/order, status history, batch, payment, feedback, consent, audit)
- Master-data seed scripts (Kenya geography, category / sub-category taxonomy, role definitions, outcome-type dictionaries per vertical)
- Migration strategy document (MySQL → future PostgreSQL)
- Data-dictionary site or Confluence page
- No application code yet — schema, seeds, and documentation only

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Phase 2: Beckn Network Onboarding

Goal

Get Kuza recognised as a BAP and a BPP on the Beckn testnet (and later production) networks.

Inputs Required

Input	Type	Provider
Beckn Testnet Subscriber Registration approval	Account	Beckn
Production-Network Subscriber Registration approval	Account	Beckn (later)
Public Domain Names (e.g., `bap-pilot.kuza.one`, `bpp-pilot.kuza.one`)	DNS records	Kuza
TLS Certificates for above domains	Certificates	Kuza (via Let's Encrypt or GCP-managed)
Subscriber IDs (one for BAP, one for BPP) — naming approved by network	Decision	Kuza + Beckn
Ed25519 Key Pairs for BAP and BPP	Generated keys	Kuza (procedure documented)
Network ID Assignment (e.g., `kuza-pilot.beckn.one/testnet`)	Decision	Beckn
Schema Hosting URLs for the 4 generic schemas	URLs	Kuza
ONIX Adapter Image Version (which release of `fidedocker/onix-adapter` to use)	Decision	Kuza tech lead
ONIX Routing YAMLs (BAPCaller, BAPReceiver, BPPCaller, BPPReceiver)	Config files	Kuza
Discovery Service Endpoint URL	URL	Beckn Fabric

Input	Type	Provider
Catalog Service Endpoint URL	URL	Beckn Fabric
DeDi Registry Endpoint URL	URL	Beckn Fabric

Expected Outputs

- ONIX-BAP and ONIX-BPP deployed on GCP (Docker on Cloud Run or GKE)
- BAP and BPP registered on DeDi Registry testnet with keys, callback URIs, subscriber IDs
- 4 generic extension schemas registered with Beckn Catalog Service
- Signed request / verified response working end-to-end in testnet (manual Postman-driven `ping` crossing the network)
- Operational runbook for registry and ONIX

Phase 3: Core Platform Services

Goal

Build the Flow Engine, BAP, and BPP — the platform "heart".

Inputs Required

Input	Type	Provider
Phase 1 outputs (data model live)	Dependency	Kuza
Phase 2 outputs (ONIX deployed and registered)	Dependency	Kuza
Flow Template Spec — YAML schema for declaring a flow template	Decision	Kuza tech lead
Initial Flow Templates List (which templates needed for	List	Kuza product team

Input	Type	Provider
pilot, naming convention)		
Cascade Rules Catalogue (which outcome triggers which cascade — text spec)	List per vertical	Kuza + domain SMEs
Batch Rules Catalogue (when batches are created, who can close them)	List	Kuza product team
SLA Targets (max latency for `discover` / `confirm`; max queue depth)	Decision	Kuza tech lead
Capacity Sizing Estimate (peak concurrent flows for pilot)	Estimate	Kuza ops
Internal API Naming Convention (Channel Handler ↔ Flow Engine ↔ BAP / BPP)	Decision	Kuza tech lead
Logging / Observability Standard (log format, trace IDs, metrics taxonomy)	Decision	Kuza tech lead

Expected Outputs

- Flow Engine service: 7-step state machine, flow-template loader, initiation-mode handling, cascade logic, batch management
- BAP service: receives requests from Channel Handlers, constructs Beckn messages, handles async callbacks
- BPP service: receives Beckn messages from ONIX, queries platform for provider catalog, constructs responses
- Unit tests for state transitions, cascade rules, batch lifecycle
- Internal API contracts (Channel → Flow Engine → BAP; Platform → BPP)
- Flow Engine admin tool (read-only, for support / ops)

Phase 4: Channel Integration — WhatsApp, Web App, Google AI

Goal

Connect the platform to the user-facing surfaces (WhatsApp, Provider Web App, Google AI).

Inputs Required

Input	Type	Provider
WhatsApp Business Account ID + API token	Credential	Kuza (Meta-issued)
Trigger Word List (which words / phrases route to which handler)	List	Kuza product team
Read-only access to existing `webhook.php` source	Repo access	Kuza platform team
WhatsApp Outbound Template Content — drafted in EN + SW for each notification scenario	Content (table)	Kuza product team + content writer
WhatsApp Template Approval from Meta (per template)	Approval	Meta (via Kuza)
Provider Web App UI Mockups (catalog mgmt, order list, order detail, batch view, profile)	Mockups	Kuza UX
Brand Assets (logo, colour palette, typography, iconography)	Asset bundle	Kuza brand team
Web App Authentication Mechanism (WhatsApp link	Decision	Kuza product + security

Input	Type	Provider
with token / OTP / phone-based)		
Google Content API Endpoint URLs + credentials	Credentials + URLs	Google
Google AI Interface (Vertex AI / Gemini) credentials + model selection	Credentials + decision	Google + Kuza tech lead
Location Normalisation Acceptance Criteria (when does AI output get auto-confirmed vs ask user)	Decision	Kuza product team
Messaging Queue Tooling Decision (Pub/Sub vs Cloud Tasks vs cron)	Decision	Kuza tech lead
Provider Notification SLA (how fast after order confirmed)	Decision	Kuza product team

Expected Outputs

- Channel Webhook router (PHP module in the existing platform) with trigger-word / message-ID-based branching
- Domain-specific Flow Handlers (Local Services handler, Health Services handler) — PHP modules, each consuming Flow Engine APIs
- Provider outbound notification worker (messaging queue + cron) decoupled from the inbound webhook
- Kuza Web App v1 — responsive, no native install — with:
  - Provider onboarding and catalog management
  - Order / flow inbox + details views
  - Role-specific views (Facilitator / Provider / Coordinator)
- Google AI integration library: location normalisation (from Channel Webhook), Content API adapter (from BPP for Advisory)
- WhatsApp template library (registered with Meta) for all outbound patterns

Phase 5: Vertical Rollout — Local Services Domain

Goal

Ship the 4 local-services flows end-to-end, covering the `SEEKER_INITIATED` pattern.

Inputs Required

Input	Type	Provider
Provider Onboarding List — agripreneurs (name, phone, type, service area, services offered, pricing, capacity, availability)	Dataset (CSV per provider type)	Kuza field team
Service Catalog Per Provider (services this provider sells, at what unit prices)	Dataset	Kuza field team
Service Area Mapping (which provider serves which county / sub-county)	Dataset	Kuza field team
Buyer / Aggregator List for Market-Produce (name, location, what they buy, buying prices, collection mode)	Dataset	Kuza field team
Vertical Configuration per local-services vertical (default values, dropdown content)	Per-vertical config	Kuza product team
Predefined Feedback Options Lists (per category)	Lists	Kuza product team
Cancellation Policy Defaults per vertical (window hours, reasons enum)	Decision per vertical	Kuza product team

Input	Type	Provider
Sample Seeker (Farmer) Profile Set (≥10 test farmers across pilot geography)	Dataset	Kuza field team
Vertical-Specific WhatsApp Conversation Scripts (numbered text + button labels in EN + SW)	Per-vertical script	Kuza product team + content writer
Test Cases Per Vertical (happy-path + edge cases)	Test plan	Kuza QA

Expected Outputs

- Flow templates defined (YAML / config) for each vertical:
  - `agri-livelihood-poultry` (vaccination, feed, brooding, incubation, etc.)
  - `household-education`, `household-health`
  - `market-livestock-products` (reversed economics)
  - `advisory-agri`
- `verticalAttributes` schemas documented per vertical
- Vertical-specific flow-handler logic in the Local Services Flow Handler
- WhatsApp message content (farmer conversations, provider notifications)
- End-to-end integration tests per vertical
- Pilot-ready for the 4 local-services flows

Phase 6: Vertical Rollout — Health Services Domain

Goal

Ship the health flows, introducing the `PROVIDER_INITIATED`, `SYSTEM_INITIATED`, cascade, and batch primitives.

Inputs Required

Input	Type	Provider
Health Facility Master List (name, type, location, capabilities, operating hours, credential ID)	Dataset (CSV)	Health partner / disease-programme
Facility Capability Tags per Facility (GeneXpert, X-ray, Spirometry, DRTB-certified, etc.)	Mapping	Health partner
CHP Roster (name, phone, assigned villages / sub-county, credential ID, supervisor)	Dataset	Health partner
SCTLC / CTLC Roster (name, phone, assigned geography, supervisor, payment-claim ID)	Dataset	Health partner
Clinical Staff Roster per Facility (clinician, nurse, lab, pharmacy — name, role, credential)	Dataset	Health partner
Disease-Programme Register Integration Credentials (read + write API access)	Credentials + endpoint URLs	Health partner
Patient UUID Generation / Reconciliation Rules (how UUIDs map to register IDs)	Decision	Health partner + Kuza
Symptom-Screening Checklist Content (questions, scoring rules) per disease	Per-disease content	Health partner clinical team
Contact-Tracing Rules — which contact relationships	Rules per disease	Health partner clinical team

Input	Type	Provider
trigger cascades, exposure window defaults		
Cascade Triggers per Outcome (e.g., TB_POSITIVE → which cascades fire)	Mapping	Health partner + Kuza
Outbound Notification Templates for Passive Seekers (contact-tracing, post-TB recall) — EN + SW + content-team-reviewed for stigma mitigation	Content	Kuza content + health-partner sign-off
Consent Text Content (DPDP / data-sharing wording) — EN + SW	Content	Health partner legal + content writer
Campaign Plan Templates (CTLC's view of how to plan an ACF campaign)	Template	Kuza product + health partner
Clinical SLAs (lab turnaround, follow-up windows)	Per-disease decisions	Health partner
Test Cases per Health Vertical including cascade scenarios	Test plan	Kuza QA + health-partner clinical reviewer

Expected Outputs

- Flow templates for each health vertical:
  - `health-tb-screening` (CHP-facilitated primary path)
  - `health-tb-acf` (campaign-style)
  - `health-tb-contact-tracing` (cascaded)
  - `health-tb-treatment`
  - `health-tb-supervision`
  - `advisory-health-tb`
- CHP-specific Web App views (batch start / close, household visit list, symptom checklist, referral creation)

- Cascade engine rules loaded (positive TB → treatment flow + contact-tracing batch)
- Batch-management UI (CHP daily batch, CTLC campaign batch)
- Outbound WhatsApp notification templates for passive seekers
- Disease-programme register bidirectional sync (read on `discover`, write on `record`)
- End-to-end integration tests per flow, including cascade scenarios
- Pilot-ready for the 6 health-services flows

Phase 7: Pilot Launch

Goal

Put real households, facilitators, and providers on the system.

Inputs Required

Input	Type	Provider
Pilot Geography Confirmation (which Counties + Sub-Counties for go-live)	Decision	Steering Committee
Pilot Cohort Targets (N farmers, N households, N CHPs, N facilities, N agripreneurs)	Number per category	Steering Committee
Onboarding Schedule (per cohort, by date)	Calendar	Kuza pilot manager
Training Material — Facilitator Guide	Document (slide deck + quick-reference)	Kuza training team
Training Material — Provider Guide	Document	Kuza training team
Training Material — Coordinator Guide	Document	Kuza training team

Input	Type	Provider
Training Trainer Roster (who delivers training, where, when)	List + schedule	Kuza pilot manager
Support Team Roster + Escalation Path	List	Kuza pilot manager
Go / No-Go Checklist (sign-offs needed before launch)	Checklist	Kuza PM
Pilot Communication Plan (announcements, partner messaging, expectation-setting)	Plan	Kuza + partners
Monitoring Dashboard URLs (for ops visibility on day 1)	URLs	Kuza ops
Incident Response Playbook	Playbook	Kuza ops

Expected Outputs

- Training completed for Facilitators, Providers, Coordinators
- Providers onboarded with catalog (services, pricing, service areas, capabilities)
- Farmers / households registered (Seeker profiles)
- Monitoring dashboards live (flow volumes, error rates, latency)
- Support channel active (WhatsApp / phone) for early-user issues
- Go-live communication sent to pilot cohort
- First live flows executed and closed end-to-end across all domains

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Phase 8: Stabilisation & Analytics

Goal

Move from “it works” to “we know how well it works and can improve it.”

Inputs Required

Input	Type	Provider
KPI List with Definitions (conversion rate, lead time, outcome distribution per vertical, etc.)	Document	Kuza product team
Pilot Evaluation Criteria (what defines pilot success)	Document	Steering Committee
Analytics Tooling Decision (Looker / Data Studio / custom)	Decision	Kuza tech lead
Data Access Roles (who sees what)	Matrix	Kuza + partners
Reporting Cadence (weekly ops report, monthly steering report)	Schedule	Kuza PM
Feedback Collection Mechanism for Pilot Users	Decision (survey / interview / in-app)	Kuza product team

Expected Outputs

- Descriptive analytics dashboards (Phase 1 of the Analytics Roadmap): flow conversion funnels, provider performance, facilitator productivity, outcome rates per vertical, geographic heat-maps
- Data-quality monitoring (completeness, referential integrity, cascade closure rates)
- Operational KPIs defined and tracked
- Feedback loop into product backlog (what to fix, what to extend)
- Foundation laid for later predictive and surveillance analytics (not built yet — data is being collected correctly)

Critical-Path Items

Inputs that most often delay pilots of this shape. Flagging them here so owners can start these earlier than the phase in which they are strictly required:

1. **Master-data lists** (Phase 1) — geography is usually fine; **vertical taxonomies and sub-categories** can stall waiting on SMEs. Start SME conversations during Phase 0.
2. **WhatsApp template approvals** (Phase 4) — Meta's approval can take days per template; batch-submit templates early, even if Phase 4 has not formally begun.
3. **Facility capability + CHP roster data** (Phase 6) — these usually live in spreadsheets across partner organisations and need reconciliation. Kick off data-cleanup with the health partner during Phases 1–2.
4. **Predefined feedback-option lists per vertical** (Phase 5) — easy to defer, hard to retrofit. Capture these during Phase 1 master-data work if possible.
5. **Cascade trigger rules** (Phase 6) — require clinical sign-off and often get stuck in review. Start clinical review conversations during Phase 3 even though the rules are consumed in Phase 6.

Next Steps

1. Review this high-level plan with the steering committee.
2. Confirm phase ordering, scope, and the critical-path items.
3. Upon sign-off, move to **detailed planning**, breaking each phase into tasks, owners, durations, and dependencies (task-level Gantt / programme plan).
4. In parallel, kick off the critical-path items flagged above so they do not block the phases in which they are formally required.

End of document.